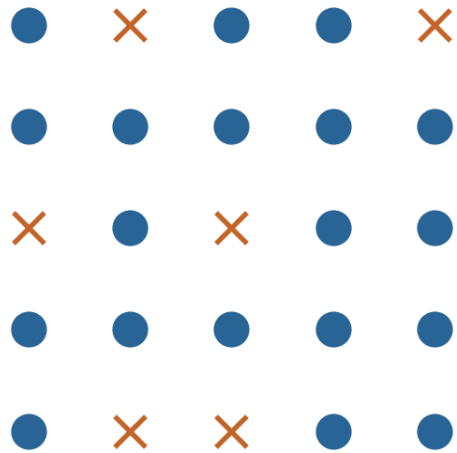


4 Measurement and Data Curation

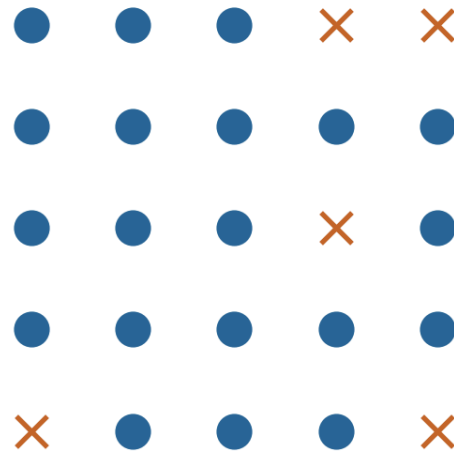
Statistical Estimation

Synthetic example: a fixed classifier has population accuracy $p = 0.8$.

Test set 1: 19/25 = 76%



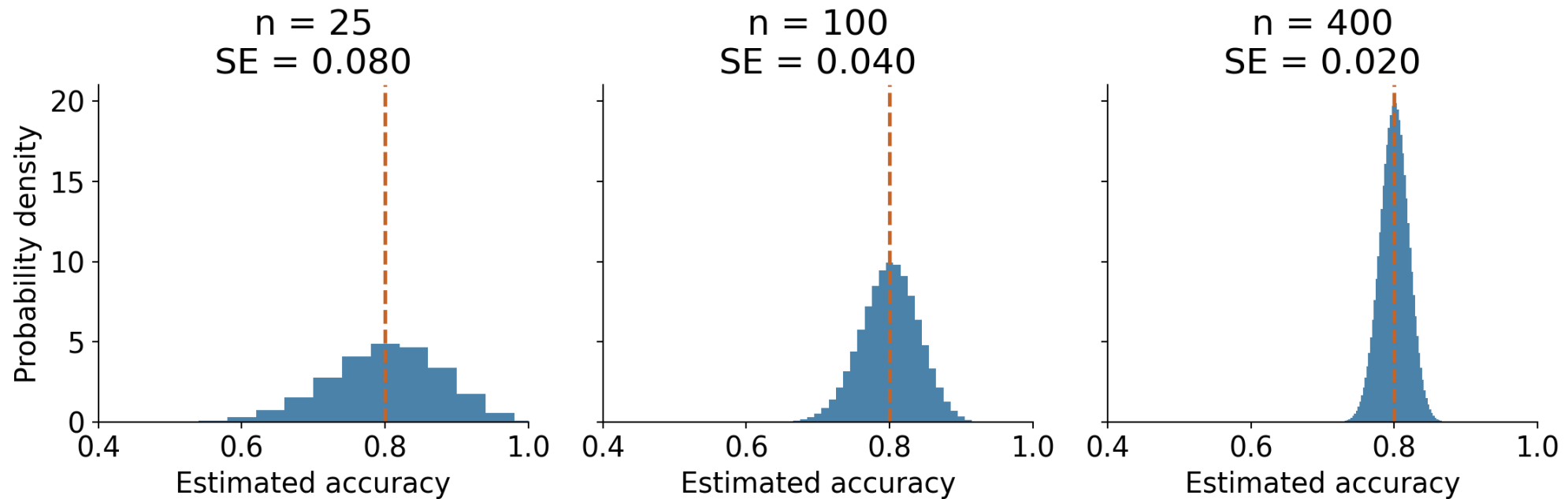
Test set 2: 20/25 = 80%



Correct prediction ● Incorrect prediction ×

$$\hat{p} = \frac{\text{number of correct predictions}}{\text{number of test examples}}.$$

Statistical Estimation



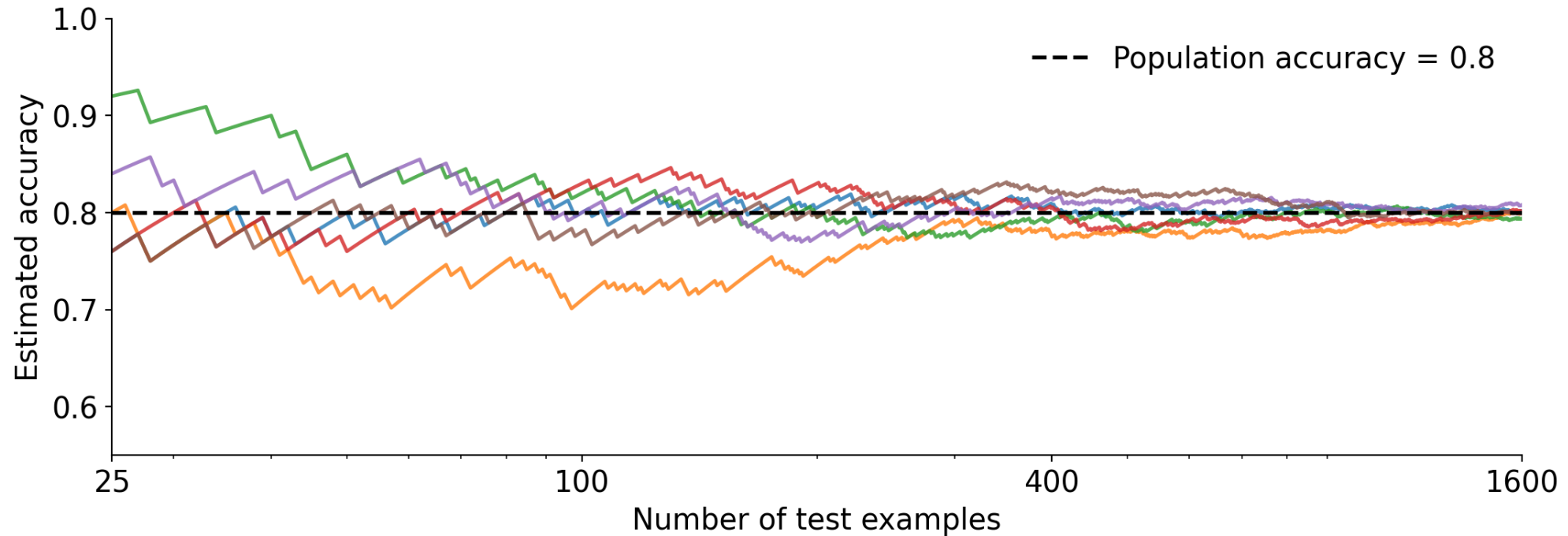
$$\mathbb{E}[\hat{p}] = p, \quad \underbrace{\sqrt{\text{Var}(\hat{p})}}_{\text{standard error (SE)}} = \sqrt{\frac{p(1-p)}{n}}.$$

4x data
1/2 std

How many times more examples do we need to halve the standard error? 🤔

Answer in class.

Statistical Estimation



More data improves precision **on average**. An individual update can move the estimate farther from p .

Model complexity

empirical risk

true risk

For a fixed finite model class and *i.i.d.* examples, with probability at least $1 - \delta$:

$$\sup_{h \in \mathcal{H}} |\underbrace{R(h)}_{\text{population error}} - \underbrace{\hat{R}(h)}_{\text{sample error}}| \leq \sqrt{\frac{\log(2|\mathcal{H}|/\delta)}{2n}}.$$

$h \in \mathcal{H}$: candidate model; n : sample size; δ : failure probability.

$R(h)$: population error; $\hat{R}(h)$: sample error (0–1 loss).

hypothesis class

Do you recognize this result? 🤔

Answer in class.

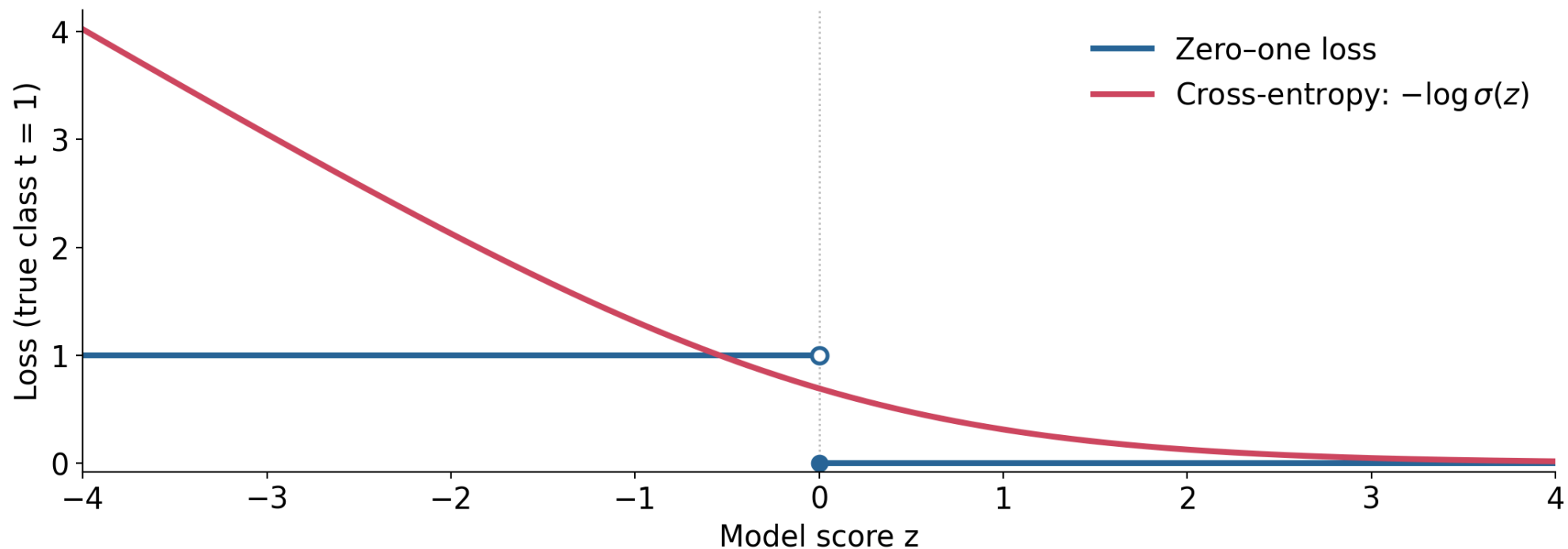
Finite-class uniform convergence bound

- More candidate models require more data for the same guarantee.
- **Regularization** restricts or penalizes model complexity.

Shalev-Shwartz and Ben-David (2014), §4.2, Corollary 4.6

Surrogate losses

Example: true class $t = 1$; predicted probability $y = \sigma(z)$.



Evaluate: accuracy. **Optimize:** a smooth surrogate such as cross-entropy.

History of Statistics

Statistik → descriptions of the **state** → **statistics**



John Sinclair

Statistical Account of Scotland (1791–1799)

938 parishes · 171 questions

Population, work, resources, and living conditions; an inquiry into opportunities for social improvement.

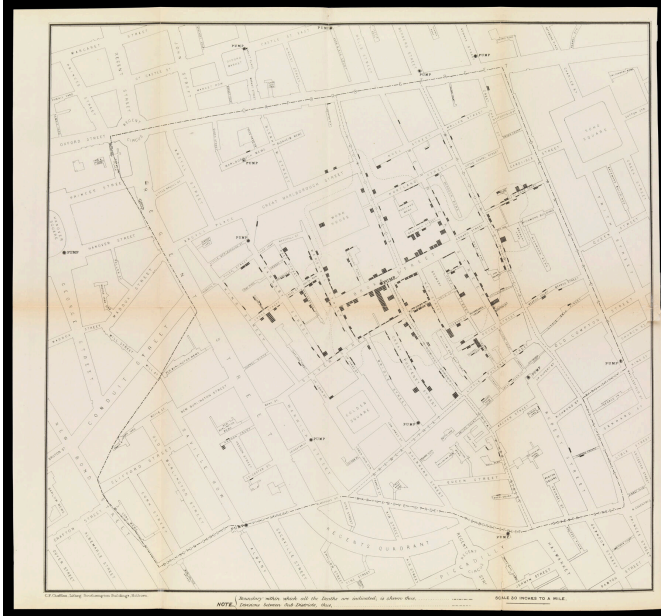
Mathematical statistics: mathematical methods for analyzing data and drawing inferences.

- **Legendre (1805), Gauss (1809):** least squares for astronomical observations and orbit estimation.
- **Galton (1886):** regression in studies of heredity.

Legendre, 1805 · Galton, 1886

[Statistical Accounts of Scotland, University of Edinburgh](#) · [Etymology](#)

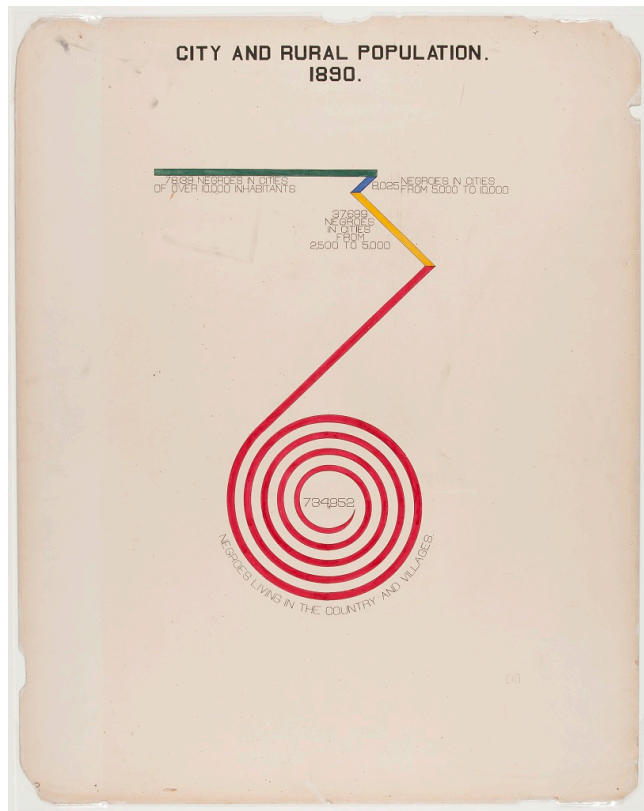
Cholera in London, 1854



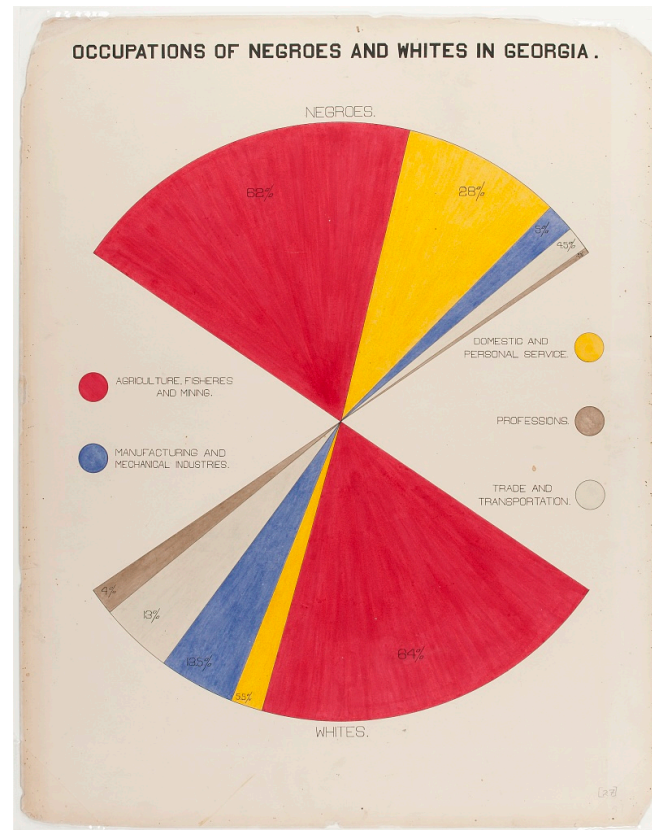
Black bars: deaths. Pump locations: potential water sources.

Snow, 1855 · Wellcome Collection

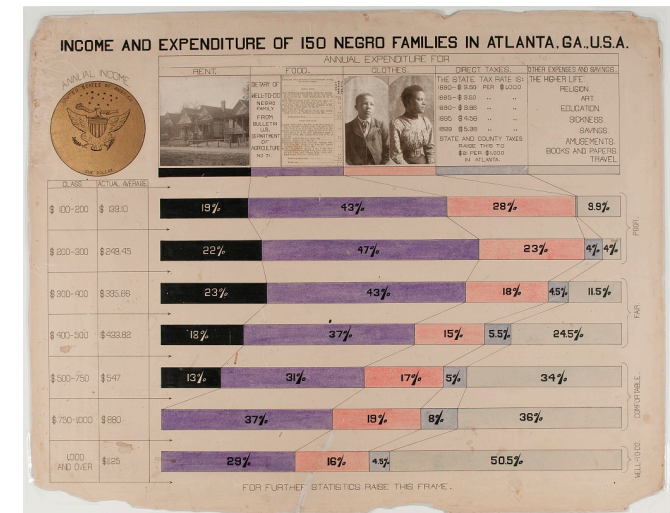
W. E. B. Du Bois — Paris Exposition, 1900



City and rural population



Occupations in Georgia

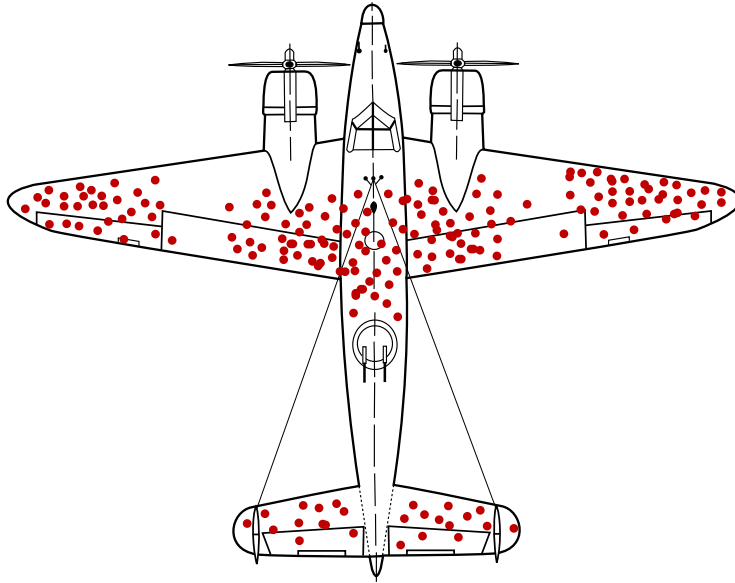


Income and expenditure

Data Confounds

$$\hat{p}(y|T)$$

Survivorship bias

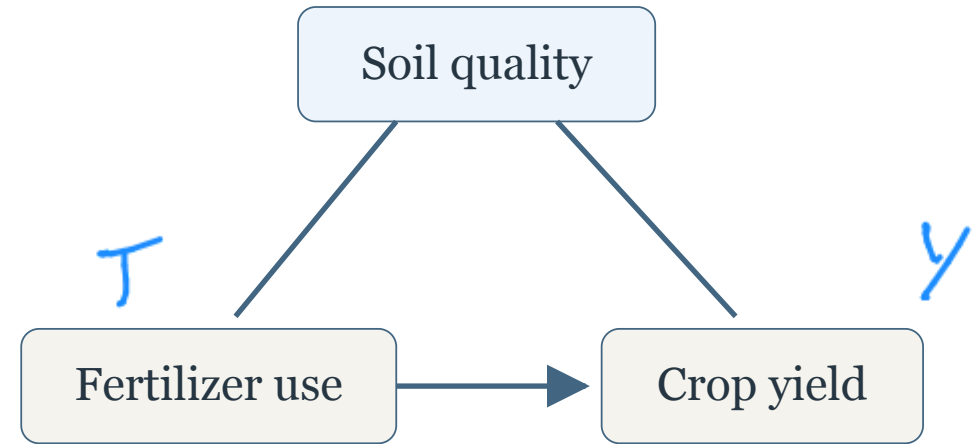


We observe damage only on aircraft that **returned**.

The sample depends on survival.

Aircraft: McGeddon / Martin Grandjean · CC BY-SA 4.0

Causal confounding



Assumed causal relationships

Soil quality affects both **fertilizer use** and **crop yield**.

The observed association mixes the fertilizer effect with soil differences.

Experimental design

Rothamsted, from 1919: Fisher's work on agricultural experiments.

Block 1	C	D	B	A
Block 2	B	D	A	C
Block 3	C	A	B	D
Block 4	D	C	A	B

A-D: four treatments; one of each in every block

Example: A = no fertilizer; B = nitrogen; C = phosphorus; D = both. Measure crop yield.

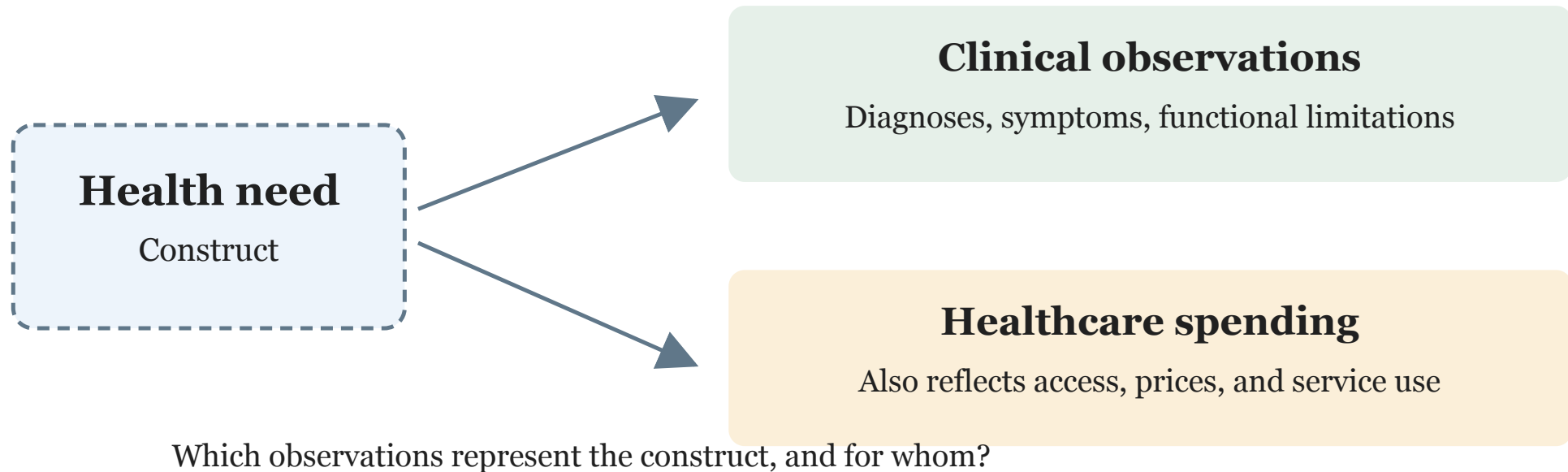
Block: a group of plots with similar soil conditions. Each block receives all four treatments.

Why include every treatment in each block, then randomize their positions? 🤔

Answer in class.

Rothamsted Research: history of statistical methods

Measurement theory



Measurement theory: how can we use observations to measure something we want to understand, such as health need or job ability?

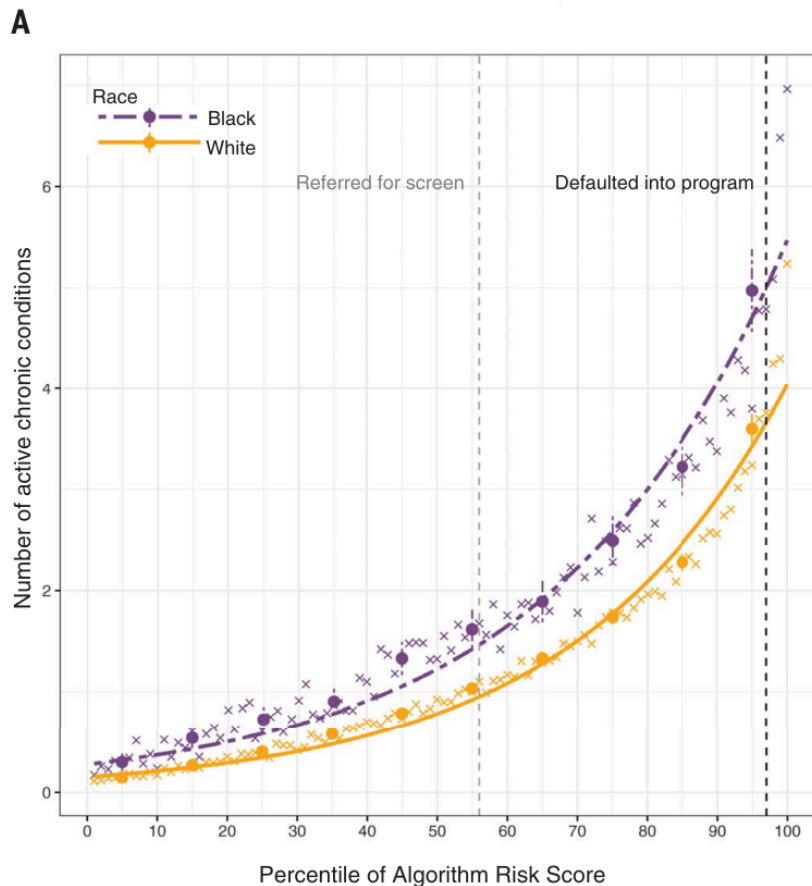
The concept we want to measure is called a **construct**.

Measure theory: the mathematics of measures, integrals, and probability.

Jacobs and Wallach (2021)

Targets and proxies

Health need → Recorded healthcare cost → Prediction loss



- **Construct:** what we want to understand or act on.
- **Measurement:** the observations used to represent it.
- **Objective:** what the learning algorithm optimizes.

More data can improve prediction of a poorly chosen proxy.

Jacobs and Wallach (2021); Obermeyer et al. (2019)

Construct validity

Construct: the concept we want to measure, such as health need or job ability.

Construct validity: the extent to which evidence and theory support interpreting a measurement as representing that construct, for its intended use.

Reliability: consistency across comparable measurements. Consistency alone does not establish validity.

Health need → Healthcare spending

Spending records can be accurate while missing unmet need among people with limited access to care.

Job ability → Timed coding score

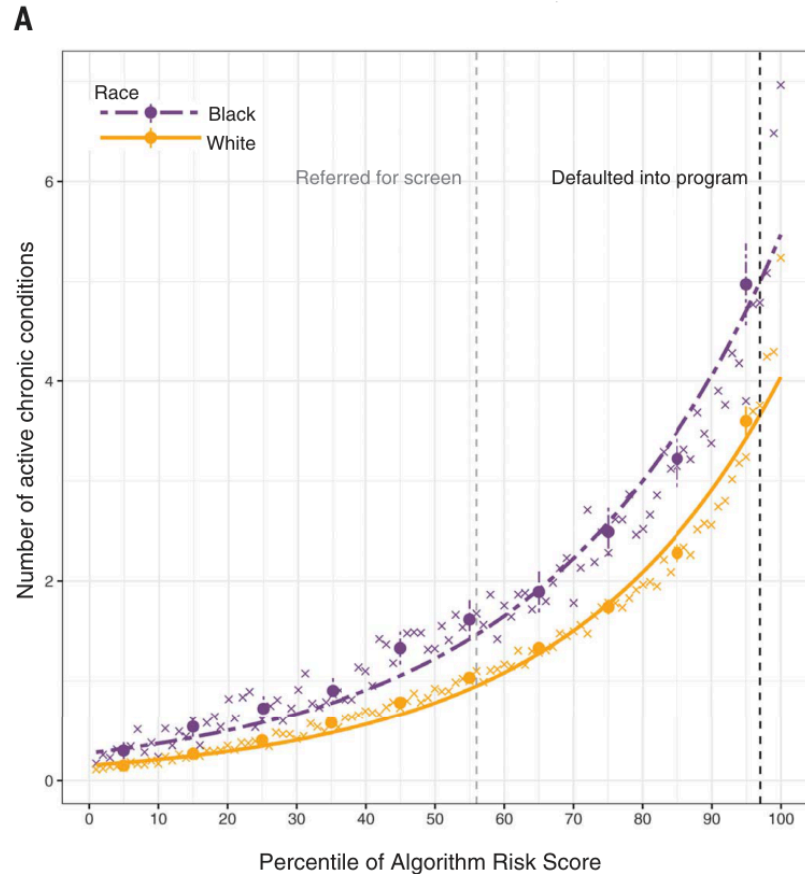
Scores may be consistent but emphasize speed while missing debugging, code review, and collaboration.

Evidence matters: do measurements cover the relevant aspects of the construct, and behave as expected across people and settings?

Jacobs and Wallach (2021); Obermeyer et al. (2019)

Measurement error and fairness

Healthcare allocation



X_{t-1} : prior-year claims, diagnoses, medications, demographics, and costs.

$Y_t = C_t$: next-year healthcare spending (training target).

$R = f(X_{t-1})$: predicted cost score used to select patients for extra care.

Intended construct: health need.

At the same score, Black patients had **more chronic illnesses**.

Obermeyer et al. (2019), Figure 1A

Measurement error and fairness

COMPAS



Bernard Parker (left): **10/high**. Dylan Fugett (right): **3/low**. Both faced drug-possession charges; their prior offences differed.

[ProPublica, 2016 · Data and analysis](#)

COMPAS: a commercial recidivism risk score.

ProPublica's data: Broward County defendants; demographics, criminal history, scores, and subsequent records.

X : recorded characteristics and criminal history.

R : COMPAS risk score.

Y : recorded rearrest within two years.

Y^* : reoffending—the outcome of interest.

Measurement error and fairness

Rearrest and reoffending

Fairness Evaluation in Presence of Biased Noisy Labels

Riccardo Fogliato
Carnegie Mellon University
Partnership on AI

Max G'Sell
Carnegie Mellon University

Alexandra Chouldechova
Carnegie Mellon University
Partnership on AI

Reoffending Y^* $\rightarrow$ Detection and enforcement $\rightarrow$ Recorded rearrest Y

- The probability that an offense produces a recorded rearrest can differ across groups.
- Fairness measured against Y need not match fairness measured against Y^* .

Sensitivity analysis: vary assumptions about unrecorded offenses and check how the fairness conclusions change.

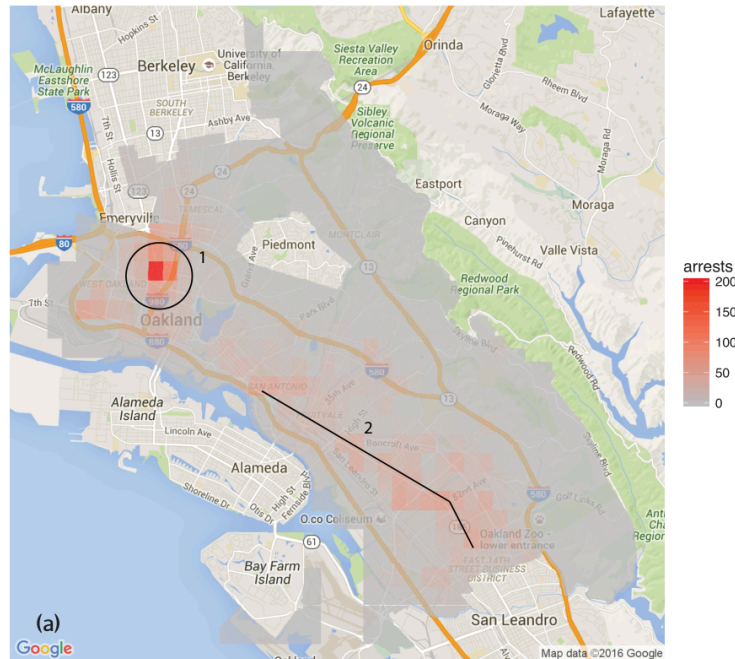
Fogliato, Chouldechova, and G'Sell (2020)

Measurement error and fairness

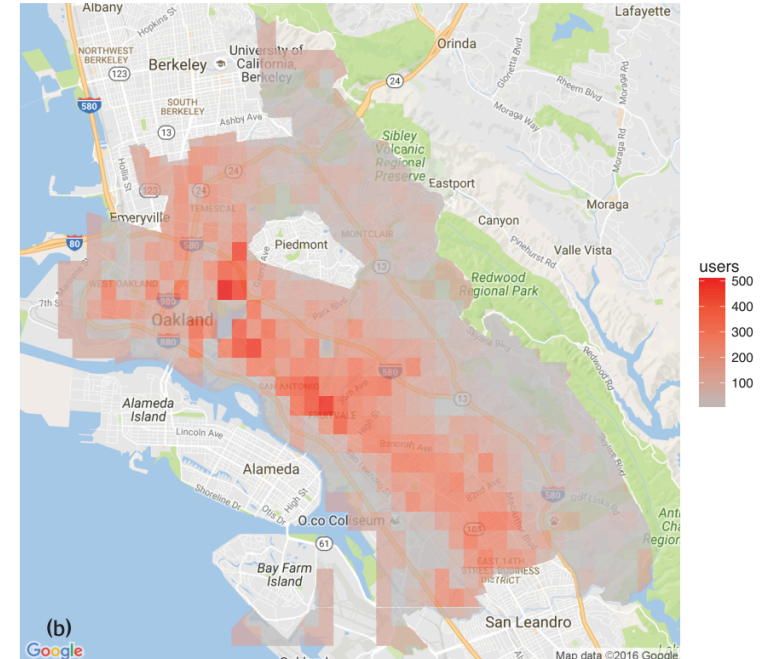
Predictive policing

Oakland, California

Recorded drug arrests (2010)



Estimated drug users (2011 survey)



Arrest records reflect **police activity** as well as drug use.

Lum and Isaac (2016)

Goodhart's law

AI benchmarks

citat

lect

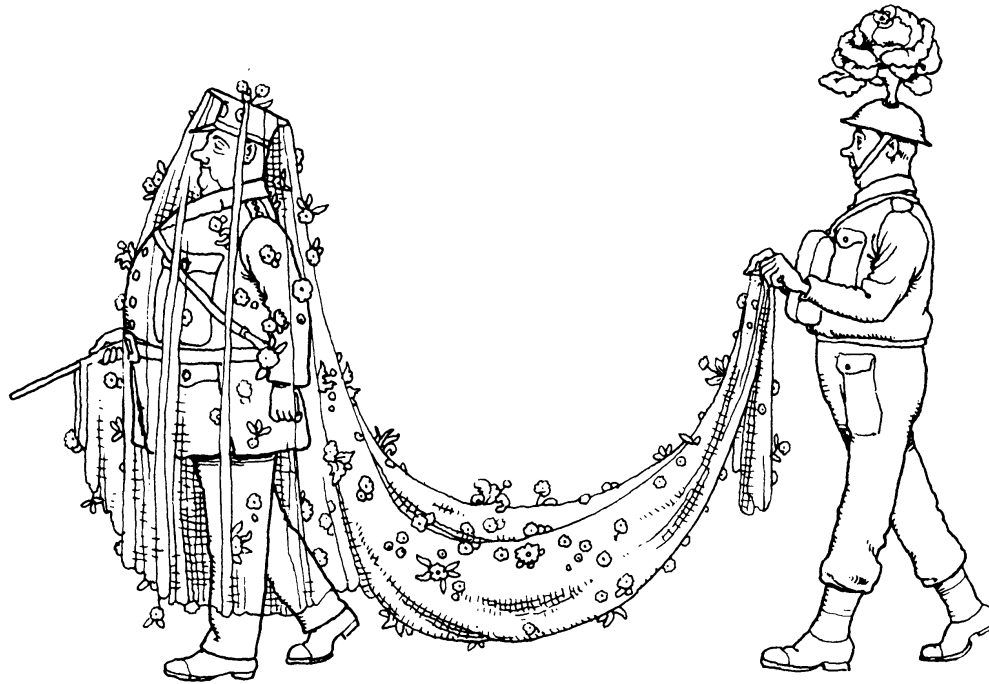
for

rec

A statistical relationship can change when a measure becomes a policy target.

What other examples of Goodhart's law can you think of? 🤔

Answer in class.

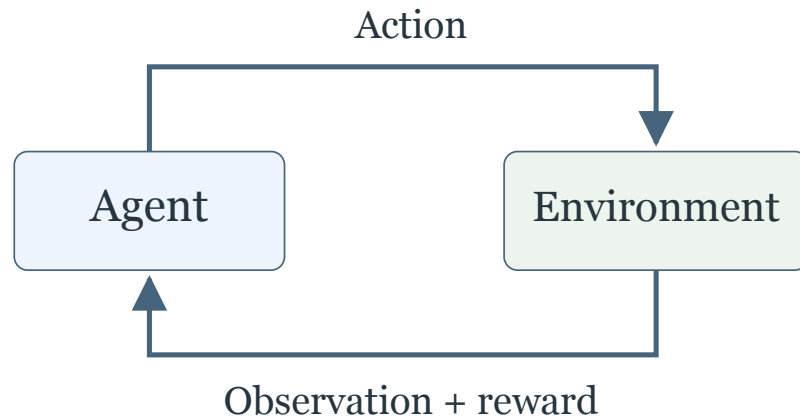


The Camouflage Expert

Goodhart, 1992 · William Heath Robinson, The Camouflage Expert · Image via Dan MacKinlay

Reward hacking

Reward hacking: increasing the specified reward without achieving the intended goal.



Goal: finish the boat race quickly.

Reward: points for hitting targets.



The agent circles through respawning targets instead of completing the course.

Clark and Amodei, 2016 · Animation

Data in an engineering system

Collection → Annotation → Training → Deployment

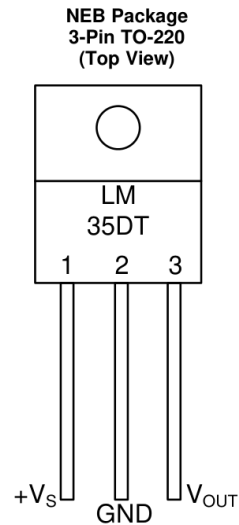
Each handoff depends on assumptions about the data.

- Population and collection conditions.
- Meaning of labels and missing values.
- Changes in context or intended use.

Sambasivan et al. (2021)

Component documentation

LM35 sensor



Typical response

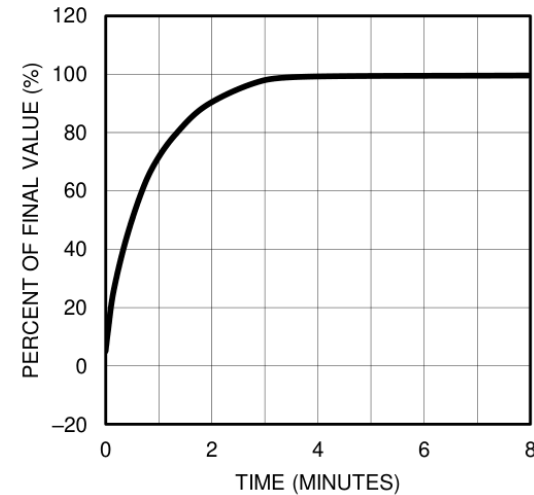


Figure 3. Thermal Response In Still Air

Motivation

For what purpose was the dataset created? Was there a specific task in mind? Was there a specific gap that needed to be filled? Please provide a description.

The dataset was created to enable research on predicting sentiment polarity—i.e., given a piece of English text, predict whether it has a positive or negative affect—or stance—toward its topic. The dataset was created intentionally with that task in mind, focusing on movie reviews as a place where affect/sentiment is frequently expressed.¹

Operating conditions

6.3 Recommended Operating Conditions

over operating free-air temperature range (unless otherwise noted)

		MIN	MAX	UNIT
Specified operating temperature: T_{MIN} to T_{MAX}	LM35, LM35A	-55	150	°C
	LM35C, LM35CA	-40	110	
	LM35D	0	100	
Supply Voltage ($+V_S$)		4	30	V

Dataset: purpose and scope

What was it designed for? Under which conditions is it useful?

Texas Instruments, LM35, §§5, 6.3, 6.9 · (Gebru et al. 2021, Appendix example, p. 15)

Dataset and model documentation

Artifact	What it describes
Datasheet	Dataset collection, composition, uses, and maintenance
Model card	Model uses, evaluation, and limitations
System card	A system's capabilities, risks, and mitigations
...	...

Documentation should support a decision about use.

A2: You will explore documentation frameworks by critiquing an existing artifact's documentation and proposing changes.

Gebru et al. (2021); Mitchell et al. (2019)

Model cards

Smile detector: intended use

Intended Use

- Intended to be used for fun applications, such as creating cartoon smiles on real images; augmentative applications, such as providing details for people who are blind; or assisting applications such as automatically finding smiling photos.
- Particularly intended for younger audiences.
- Not suitable for emotion detection or determining affect; smiles were annotated based on physical appearance, and not underlying emotions.

Training and evaluation data

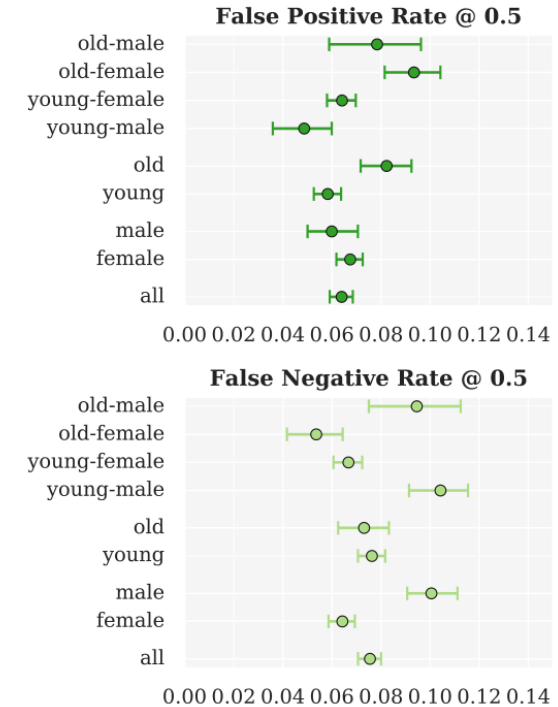
Training Data

- CelebA [36], training data split.

Evaluation Data

- CelebA [36], test data split.
- Chosen as a basic proof-of-concept.

Quantitative Analyses



Mitchell et al. (2019), Figure 2

Revising a model card

Proposed use: selecting smiling photographs of children.



What the card establishes

Smile detection, evaluated on CelebA by annotated age and gender group.

Proposed addition

“This card does not establish performance for children. Before this use, evaluate on relevant images and report errors by age and capture conditions.”

CelebA sample images

Evidence needed: suitable evaluation data, label definitions, and subgroup results.

Mitchell et al. (2019), Figure 2

Training data disclosure

2 Scope and Limitations of this Technical Report

This report focuses on the capabilities, limitations, and safety properties of GPT-4. GPT-4 is a Transformer-style model [39] pre-trained to predict the next token in a document, using both publicly available data (such as internet data) and data licensed from third-party providers. The model was then fine-tuned using Reinforcement Learning from Human Feedback (RLHF) [40]. Given both the competitive landscape and the safety implications of large-scale models like GPT-4, this report contains no further details about the architecture (including model size), hardware, training compute, dataset construction, training method, or similar.

We are committed to independent auditing of our technologies, and shared some initial steps and ideas in this area in the system card accompanying this release.² We plan to make further technical details available to additional third parties who can advise us on how to weigh the competitive and safety considerations above against the scientific value of further transparency.

The report does not provide enough detail to audit **training-data sources and coverage**.

GPT-3 reported dataset sizes and sampling weights. GPT-4 gives only broad source categories: public and licensed data.

[GPT-4 report, §2](#) · [GPT-3 paper, §2.2 and Table 2.2](#)

Documentation and accountability

What would make documentation useful evidence for a decision to deploy—or not deploy—a system? 🤔

Answer in class.

Consider the reader, the decision, and the information they can verify.

Reproducibility in machine learning

Improving Reproducibility in Machine Learning Research (*A Report from the NeurIPS 2019 Reproducibility Program*)

Joelle Pineau: ICLR 2018 Reproducibility Challenge; NeurIPS 2019 reproducibility program.

		Data	
		Same	Different
Code & Analysis	Same	Reproducible	Replicable
	Different	Robust	Generalisable

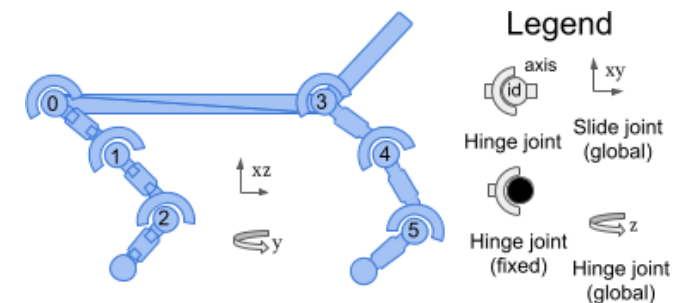
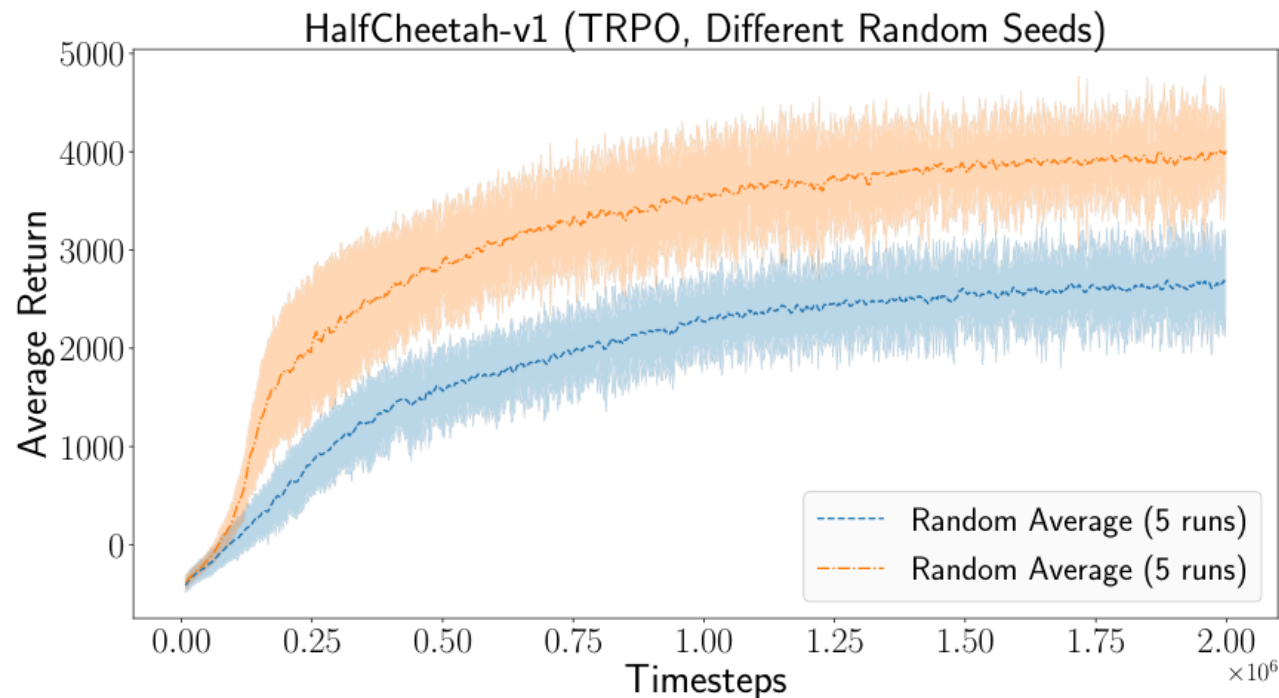
- **Code submission:** inspect and reuse implementations.
- **Reproducibility Challenge:** independent teams test reported results.
- **Reporting checklist:** document experiments and uncertainty.

Reproducing a score requires knowing **how the data and measurements were produced.**

Pineau et al. (2021)

Variation across training runs

Same algorithm and hyperparameters; different random seeds.



HalfCheetah: learn joint controls to move forward.

- Train **10 runs**, changing only the random seed; divide into **two groups of 5**.
- At each training step, plot each group's **mean return**.
- Shade **mean $\pm$ standard error**: $\bar{R} \pm s/\sqrt{5}$, with s the standard deviation across those five runs.

Henderson et al. (2018) · Figure 5 · Environment: Gymnasium

Data and evaluation research

2018–2019

Reproducibility initiatives

Can others check the reported result?

NeurIPS 2021

Datasets and Benchmarks track

Recognize data and benchmark construction as research.

NeurIPS 2026

Evaluations & Datasets track

Broader emphasis on evaluation methods and datasets.

These initiatives change **what researchers report, what reviewers examine, and what conferences recognize.**

Reproducibility, measurement validity, and usefulness in deployment remain distinct questions.

NeurIPS track announcement, March 2026

Evaluation documentation



REPRODUCIBILITY

3%

of reported scores have a complete setup recorded — the rest cannot be independently re-run.

97% have at least one undocumented field. Most often missing: max tokens (96%), temperature (94%).

ASKS

Can someone else run this evaluation and get the same number?



COMPLETENESS

76%

mean across 52,691 reported score triples.

Observed range: 7% to 93%.

ASKS

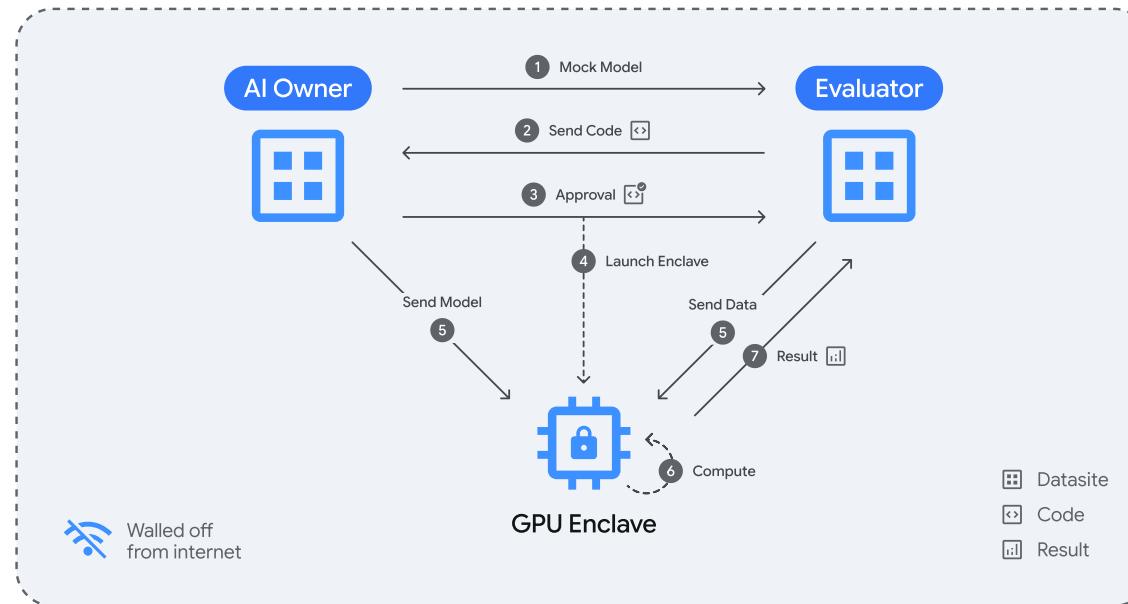
Is the benchmark itself documented well enough to interpret a score on it?

A reported score needs a description of **how it was obtained**.

[EvalEval · Quickstart illustration](#)

What's New?

Double-blind AI evaluations



Concepts: evaluator access, confidentiality, reproducibility.

What evidence would an external evaluator still need? 🤔

Answer in class.

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